

Quiz 1**Name:**

You must show work to receive credit. Unless indicated otherwise, each problem is worth equal value.

1. Let $U = \{a, b, c, d, e, f, g, h, i, j, k\}$ be the universal set, $A = \{a, b, c, d, e, f\}$ and $B = \{a, c, e, h, j\}$. Find the indicated set.

a. $A \cup B$

$$= \{a, b, c, d, e, f, h, j\}$$

b. $A \cap B$

$$= \{a, c, e\}$$

c. A'

$$= \{g, h, i, j, k\}$$

d. $(A \cap B)'$

$$= \{b, d, f, g, h, i, j, k\}$$

e. $A \cap A'$

$$= \emptyset$$

2. Let $A \subseteq U$ and $B \subseteq U$. If $n(U) = 25$, $n(A) = 10$, $n(B) = 20$, and $n(A \cap B) = 5$, find the following:

a. $n(A')$

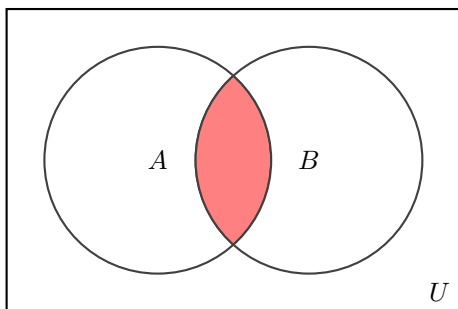
$$= n(U) - n(A) = 15$$

b. $n(A \cup B)$

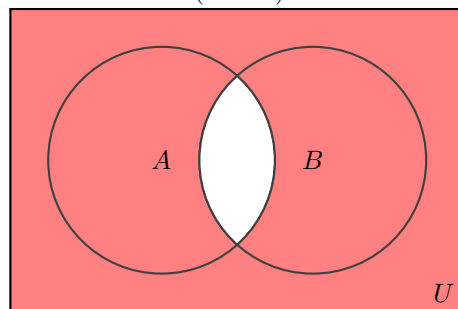
$$= n(A) + n(B) - n(A \cap B) = 25$$

3. Use the provided diagrams to shade in the region corresponding to the indicated set.

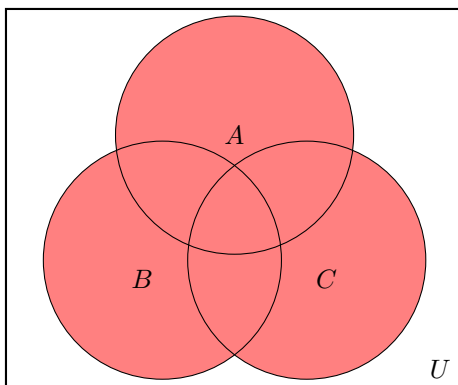
a. $A \cap B$



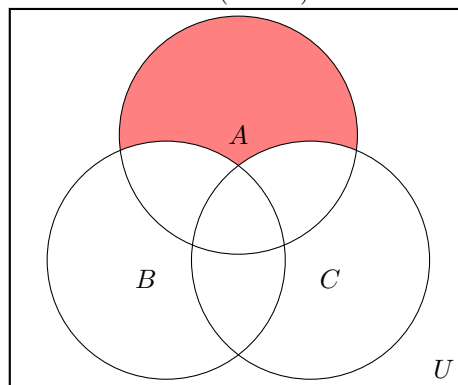
b. $(A \cap B)'$



c. $A \cup B \cup C$



d. $A \cap (B \cup C)'$



4. A group of 100 MHU students was surveyed about the type of vehicle they drive. 50 students like football, 40 students like basketball and 20 students like football and basketball. Answer the following.

- a. How many students like football or basketball?

$$\begin{aligned} n(F \cup B) &= n(F) + n(B) - n(F \cap B) \\ 70 &= 50 + 40 - 20 \end{aligned}$$

- b. How many students like football but not basketball?

$$\begin{aligned} n(F) - n(F \cap B) \\ 30 &= 50 - 20 \end{aligned}$$

- c. How many students do not like football or basketball?

$$\begin{aligned} n(U) - n(F \cup B) \\ 30 &= 100 - 70 \end{aligned}$$

5. A group of 200 mathematicians were asked about their pets. 70 had dogs, 60 had cats, 55 had fish, 25 owned all three, 10 had only cats, 10 had only dogs, and 40 had dogs and fish.

- a. How many have only cats and dogs?

20

- b. How many have only fish?

10

- c. How many have no cats, fish, or dogs?

105

